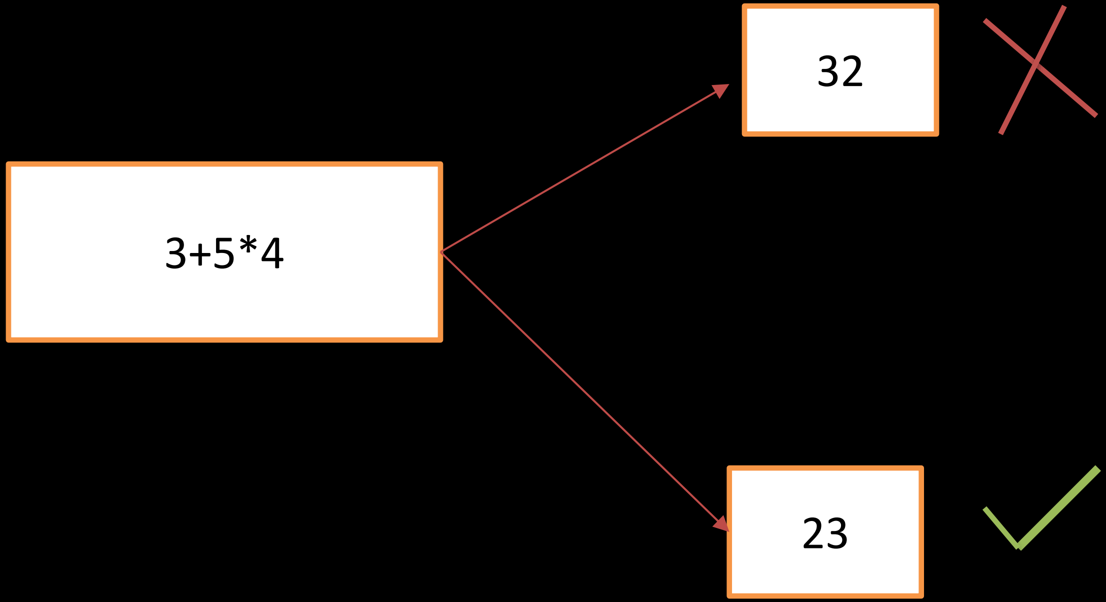
🔢 Arithmetic Operators in Programming

Operator	Meaning	Example	Result
+	Addition	4 + 2	6
-	Subtraction	4 - 2	2
*	Multiplication	4 * 2	8
/	Division	4 / 2	2
%	Modulus (Remainder of division)	5 % 2	1

❖ Arithmetic Operators(+,-,*,%,/)



Precedence order:

Operator	Description	Precedence Level	Associativity
* / %	Multiplication, Division, Modulus	Higher	Left to Right
+ -	Addition, Subtraction	Lower	Left to Right

% Modulas Operator

a % b : Remainder when a is Divided by b

14%5 =4
12%2=0
2%5=2
3%7=3

Note : Both Operand Must be int data type otherwise error ex:1.2%2===check karo

Note 1: Both Operand Must be int data type otherwise error ex:1.2%2===check karo

Note 2:Sign of result of % Operator is same as 1st Operand

Ex:-15%7=-1;

15%-7=1;

a=1+3*5 → 1+15=16

Example 1: Multiplication, Division, Modulus (*, /, % - Left to Right Associativity)

int result = 20 / 5 * 2 %
3;

Steps of Evaluation (Left to Right):

1. **Division**: 20 / 5 = 4
2. **Multiplication**: 4 * 2 = 8
3. **Modulus**: 8 % 3 = 2

Answer:

result = 2

Behaviors of Operator => Operands

5/2= 2 =>both operands are Integer type
5.0/2=2.5 =>float,int=>float
5/2.0=2.5=>int,float=>float
5.0/2.0=>float ,float=>float

```java
public class Digit {
    public static void main(String[] args) {

        int num=153;

        //print last digit

        int result1=num %10;//153%10=3
        System.out.println("last digit="+result1);
        int result2=num%100;//153%100=53;
        System.out.println("last two digit="+result2);
        int c=num/10;//153/10=15
        int result3=c%10;//15%10=5
        System.out.println(" second last digit="+result3);

    }
}
```

Use of operator:

153

Print last digit=153%10=3;

Print last two digit=153%100=53;

153

Print 2nd last digit=(153/10) %10=5;

Method 2: **Without Third Variable (Using + and - Assignment Operators)**

Swapping of number without temp variable

Swapping without a Temporary Variable

- Start
- Input two numbers, a and b
- Set a = a + b
- Set b = a - b
- Set a = a - b
- Output the swapped values of a and b
- End

Before swapping

a=10	b=20

Afther swapping

a=20	b=10

```java
public class SwapWithoutTempAddSub {
    public static void main(String[] args) {
        int a = 5, b = 10;

        System.out.println("Before Swap: a = " + a + ", b = " + b);

        a += b;  // a = a + b => a = 15
        b = a - b;  // b = 15 - 10 = 5
        a -= b;  // a = 15 - 5 = 10

        System.out.println("After Swap: a = " + a + ", b = " + b);
    }
}
```

Method 3: **Using XOR (Bitwise Assignment Operator)**

```java
public class SwapUsingXOR {
    public static void main(String[] args) {
        int a = 5, b = 10;

        System.out.println("Before Swap: a = " + a + ", b = " + b);

        a ^= b;  // a = a ^ b
        b ^= a;  // b = b ^ a
        a ^= b;  // a = a ^ b

        System.out.println("After Swap: a = " + a + ", b = " + b);
    }
}
```

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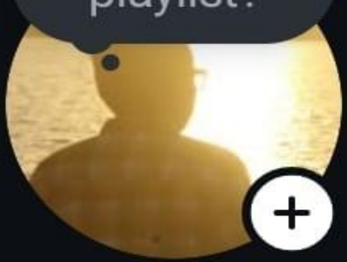

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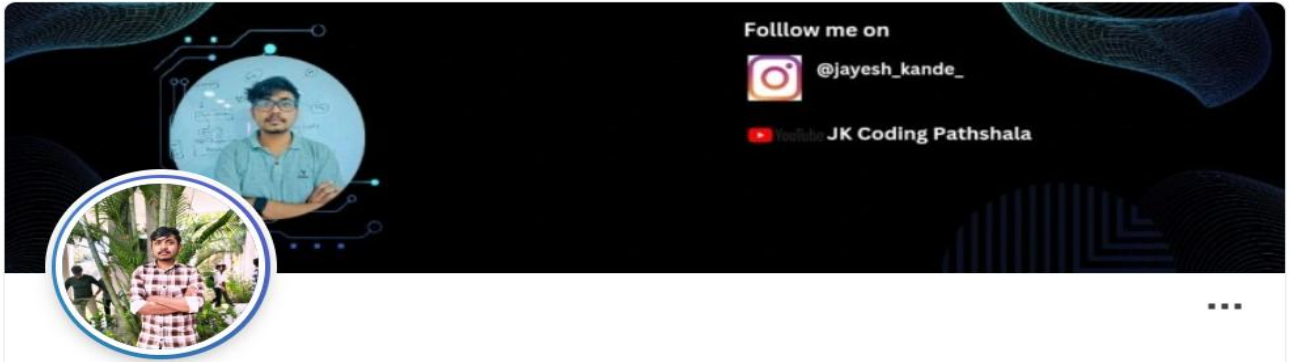

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